Question

Transition metal $\mathbf{M}$ is difficult to extract due to its dispersed presence in nature. The highest-valent oxide $\mathbf{A}$ of $\mathbf{M}$ reacts with coke and chlorine gas at high temperature to produce $\mathbf{B}$. $\mathbf{B}$ reacts with $\mathbf{M}$ to produce $\mathbf{C}$. $\mathbf{C}$ is often used to catalyze ethylene polymerization. Pyrolysis of $\mathbf{C}$ can yield $\mathbf{B}$ and $\mathbf{D}$. The crystal of $\mathbf{D}$ has a layered structure, in which all $\mathbf{M}$ are octahedrally coordinated, and all octahedra are connected to six adjacent octahedra by edge-sharing. Heating $\mathbf{D}$ can also yield $\mathbf{M}$ and $\mathbf{B}$. $\mathbf{B}$ reacts with excess ethanol to produce $\mathbf{F}$. $\mathbf{F}$ is a tetramer, and the ratio of oxygen atoms with different coordination numbers is 1:2:5. $\mathbf{B}$ reacts with excess NaCp ($\mathrm{Cp} = \mathrm{C_5H_5}$) to produce $\mathbf{G}$ or $\mathbf{H}$. Heating $\mathbf{G}$ can also produce $\mathbf{H}$. $\mathbf{G}$ satisfies the 16-electron rule, while $\mathbf{H}$ contains two bridging hydrogen atoms.

The following statements are made:

1. The minimum mass fraction of $\mathbf{M}$ in $\mathbf{A} - \mathbf{D}$ is less than 25%.

2. Ignoring carbon and hydrogen atoms, the point group of $\mathbf{F}$ is D_{2h}.

3. There are four chemically distinct types of hydrogen in $\mathbf{G}$.

4. $\mathbf{M}$ in $\mathbf{H}$ satisfies the 18-electron rule.

Then the following options contain all the correct statements:

A. All other options are incorrect

B. 1

C. 2

D. 3

E. 4

F. 1, 2

G. 1, 3

H. 1, 4

I. 2, 3

J. 2, 4

K. 3, 4

L. 1, 2, 3

M. 1, 2, 4

N. 1, 3, 4

O. 2, 3, 4

P. 1, 2, 3, 4

Answer

Correct Answer: D

Detailed Explanation

Based on the fact that $\mathbf{C}$ is often used to catalyze ethylene polymerization, it can be inferred that the core element of this question is Ti.

CHECKPOINT 1 PTS

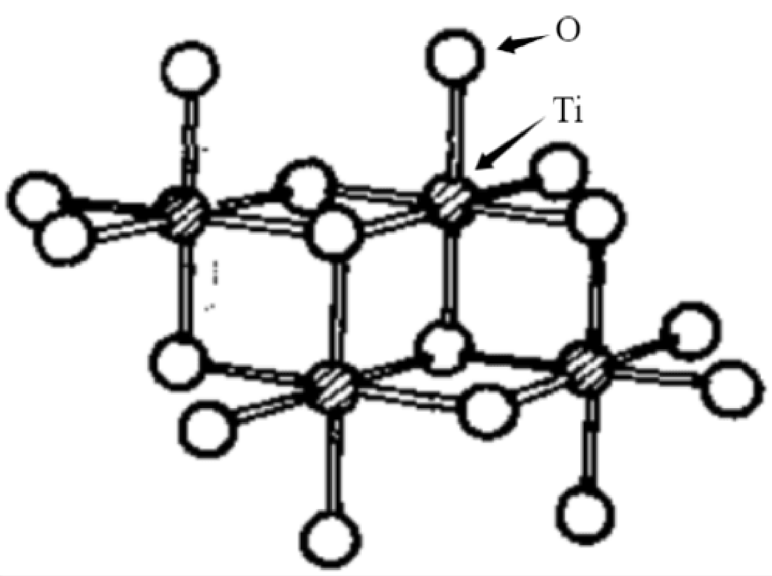
$\mathbf{M}$ is Ti

Therefore, it can be immediately obtained that $\mathbf{A}$ is $\mathrm{TiO_2}$, $\mathbf{B}$ is $\mathrm{TiCl_4}$, $\mathbf{C}$ is $\mathrm{TiCl_3}$, and $\mathbf{D}$ is $\mathrm{TiCl_2}$. Among them, the mass fraction of Ti is the smallest in $\mathrm{TiCl_4}$, which is 25.23%.

CHECKPOINT 1 PTS

The minimum mass fraction of $\mathbf{M}$ in $\mathbf{A}-\mathbf{D}$ is 25.23%

$\mathbf{B}$ reacts with excess ethanol to generate $\mathbf{F}$. $\mathbf{F}$ is a tetramer. Based on the oxygen atom ratio, it can be deduced that $\mathbf{F}$ is a tetramer of $\mathrm{Ti(C_2H_5O)_4}$. From the oxygen atom ratio, it can be deduced that the structure of $\mathbf{F}$ is (carbon and hydrogen atoms are omitted in the figure):



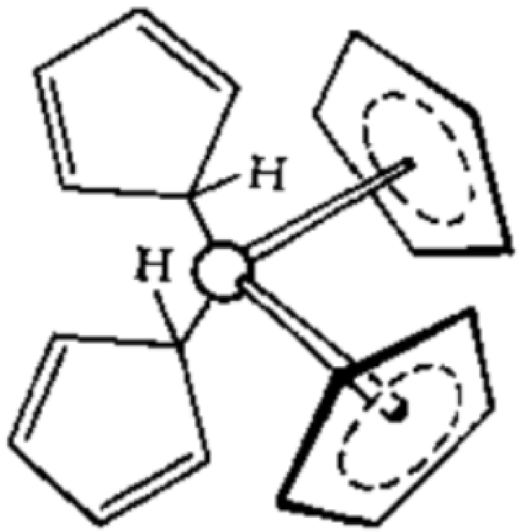
CCO1[Ti]2(OCC)(OCC)(O([Ti]3(OCC)(O4CC)(O2([Ti]15(OCC)(O([Ti]4(OCC)(OCC)(O53CC)OCC)CC)OCC)CC)OCC)CC)OCC

This structure has a C_2 axis and a mirror plane perpendicular to it, with C_{2h} symmetry.

CHECKPOINT 1 PTS

Ignoring carbon and hydrogen atoms, the point group to which $\mathbf{F}$ belongs is C_{2h}

$\mathbf{B}$ reacts with excess NaCp to generate $\mathbf{G}$ or $\mathbf{H}$. $\mathbf{G}$ should be a mononuclear complex, so it is speculated that $\mathbf{G}$ is $\mathrm{TiCp_4}$. Since $\mathbf{G}$ satisfies the 16-electron rule, it can be obtained that two Cp provide 6 electrons and two Cp provide 2 electrons. Its structure is easily obtained as:



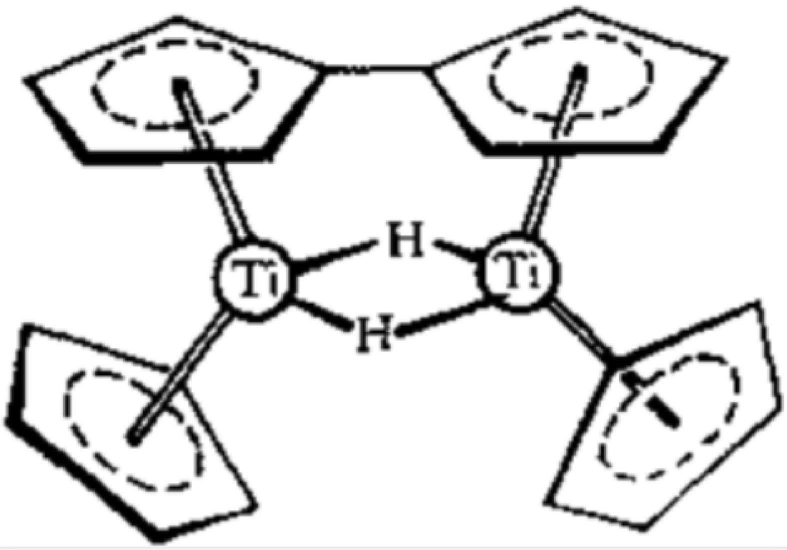
The figure shows the structure of $\mathrm{TiCp_4}$, where two Cp ligands provide 6-electron coordination and two Cp ligands provide 2-electron coordination.

There are 4 kinds of hydrogen with different chemical environments.

CHECKPOINT 1 PTS

There are four kinds of hydrogen with different chemical environments in $\mathbf{G}$.

$\mathbf{G}$ can also generate $\mathbf{H}$ upon thermal decomposition. There are two hydrogen bridge bonds in $\mathbf{H}$, so it can be inferred that a dehydrogenative coupling reaction of Cp has occurred. Therefore, the chemical formula of $\mathbf{H}$ can be obtained as $\mathrm{Ti_2(Cp-Cp)Cp_2H_2}$, and its structure is:



The figure shows the structure of $\mathrm{Ti_2(Cp-Cp)Cp_2H_2}$, where two Cp rings in $\mathrm{Cp-Cp}$ provide 6-electron coordination to the two Ti respectively to form a bridge, two Cp provide 6-electron coordination to the two Ti, and two H both connect the two Ti as bridging ligands.

The number of electrons around Ti is 17, which does not satisfy the 18-electron rule.

CHECKPOINT 1 PTS

$\mathbf{M}$ in $\mathbf{H}$ does not satisfy the 18-electron rule.